

MT209: Fundamentals of Metal Extraction

End-Semester Examination

Total Marks [40]

Date: 20th Apr 2026

Time: 10:00 to 13:00 hrs

1. Mention significant fundamental difference between concentration, reduction and refining process using an example of each. [03]
2. Differentiate the following unit processes: [10]
 - a) Selective distillation and Reactive distillation
 - b) Chlorination refining and Chloridizing roasting
 - c) Electrochemical-refining and Electrostatic concentration
 - d) Filtration and Sizing
 - e) Low pressure Pidgeon process and Low pressure vacuum degassing
3. Calculate the following: [02]
 - (a) Voltage due to electrolyte resistance and total voltage requirement for electrolytic plant with the following parameters:
 - Current density: $\beta \text{ Am}^{-2}$ [β is last three digits of Student ID]
 - Specific conductance of electrolyte = $50 \text{ ohm}^{-1}\text{m}^{-1}$
 - Separation between electrodes: 10 cm
 - Thermodynamic decomposition voltage = 1.25 V
 - Total over-potential at electrodes = 0.50 V
 - Total voltage drop from both electrode plates (anode and cathode) to the bus bars = 0.20 V
 - (b) Energy required (in kW-h) per kg of Cu (equivalent wt. = 31.8 g/mol) to be deposited using the above electrolytic plant having current efficiency of α % [α is last two digits of Student ID] [02]
 - (c) Total current per tank, if the above process with same current density is carried out using the tanks which has 11 cathodes and 10 anodes in parallel and dimension of electrode is 50 cm X 100 cm, which is entirely immersed in electrolyte. [01]
 - (d) Annual capacity of plant (in terms of Cu production per annum) with continuous operation of 24 hours a day and 365 days per annum for a plant with 2000 such tanks. [02]

$0.92 \times 1 \times 365 \times 24 \times 3600 \times 96485 \times 2$

$\frac{t \times 1000}{960 \times 10} = 2000$
4. Describe Blast Furnace process of iron making using a (i) heat diagram, mention the following: [01]
 - (ii) list all the input raw materials with composition, and temperature for each of them. [03]
 - (iii) composition and temperature of product Hot Metal. [01]
 - (iv) list all the by-products generated with composition, and temperature for each of them. [02]
 - (v) mention all the chemical reactions [03]
5. BOF Steel making process is also known as LD converter named after two towns in Austria, Linz and Donawitz. Compare LD converter with Pierce Smith Converter in terms of (i) metal being processed, (ii) significance, (iii) application, (iv) input raw materials, (v) output product, (vi) chemical reactions, (vii) temperature of operation, (viii) size, (ix) capacity, (x) design etc. Schematic diagrams of both may also be used for comparison. [10]